



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
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**LCS in cui si alternano
dispari (in posizioni dispari) e pari (in posizioni pari)**

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[Problema LCSdp(X,Y)]

INPUT:

- $X = \langle x_1, x_2, \dots, x_m \rangle$
- $Y = \langle y_1, y_2, \dots, y_n \rangle$

OUTPUT:

- **LCSdp(X,Y)**, Longest Common Subsequence in cui si alternano dispari (in posizioni dispari) e pari (in posizioni pari)
(**NB:** la sequenza vuota è una soluzione)

[Problema LCSdp(X,Y)]

ESEMPIO:

$X = \langle 1, 2, 5, 3, 5, 9, 9, 11, 18, 10, 19 \rangle$

$Y = \langle 14, 2, 5, 5, 9, 9, 11, 18, 20, 19, 10 \rangle$

$\text{LCSdp}(X,Y) = \langle 5, 18, 19 \rangle$

$\langle 2, 5, 18, 19 \rangle$ non è una $\text{LCSdp}(X,Y)$

[Sottoproblema ausiliario]

Sottoproblema ausiliario di dimensione (i, j)

Trovare $\text{LCSdp}_v(X_i, Y_j)$, cioè la LCS dei prefissi X_i e Y_j , in cui si alternano dispari (in posizioni dispari) e pari (in posizioni pari), vincolata a terminare con l'elemento x_i se $x_i = y_j$. Se $x_i \neq y_j$, allora $\text{LCSdp}_v(X_i, Y_j)$ non esiste.

$$i \in \{1, \dots, m\}$$

$$j \in \{1, \dots, n\}$$

$$c_{ij} = |\text{LCSdp}_v(X_i, Y_j)|$$

Numero totale di sottoproblemi: mn

Soluzione ottimale e valore ottimo

Soluzione ottimale

$$\text{LCSdp}(X,Y) \equiv \max\{\text{LCSdp}_v(X_i, Y_j) \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

$$\text{Valore ottimo} \equiv \max\{c_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

[Eq. ricorrenza – Casi base]

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i \neq y_j$$

$$\Rightarrow \nexists \text{ LCSdp}_v(X_i, Y_j)$$

$$\Rightarrow c_{ij} = 0$$



Eq. ricorrenza - Passo ricorsivo

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge x_i \% 2 = 0$$

$$\text{LCSdp}_v(x_i, y_j) = \langle \dots, x_p, x_i \rangle = \text{LCSdp}_v(x_p, y_q) + \langle x_i \rangle$$

$$\text{LCSdp}_v(x_p, y_q) = \max\{\text{LCSdp}_v(x_h, y_k) \mid 1 \leq h < i, 1 \leq k < j, \mid \text{LCSdp}_v(x_h, y_k) \mid \% 2 = 1\}$$

$$\max\{\emptyset\} \Rightarrow \nexists \text{LCSdp}_v(x_p, y_q) \Rightarrow \nexists \text{LCSdp}_v(x_i, y_i)$$

$$c_{ij} = c_{pq} + 1$$

$$c_{pq} = \max\{c_{hk} > 0 \mid 1 \leq h < i, 1 \leq k < j, c_{hk} \% 2 = 1\}$$

$$c_{ij} = \max\{c_{hk} > 0 \mid 1 \leq h < i, 1 \leq k < j, c_{hk} \% 2 = 1\} + 1$$

$$\max\{\emptyset\} \Rightarrow c_{ij} = 0$$



Eq. ricorrenza - Passo ricorsivo

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge x_i \% 2 = 1$$

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$$\max\{\emptyset\} \Rightarrow \nexists \text{LCSdp}_v(x_p, y_q) \Rightarrow \text{LCSdp}_v(x_i, y_j) = \langle x_i \rangle$$

$$c_{ij} = c_{pq} + 1$$

$$c_{pq} = \max\{c_{hk} > 0 \mid 1 \leq h < i, 1 \leq k < j, c_{hk} \% 2 = 0\}$$

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$$\max\{\emptyset\} \Rightarrow c_{ij} = 0$$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge x_i \% 2 = 1$$

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